

**From Person-Specific Dynamics to Population Inference: A Computationally Efficient
Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel
Dynamic Models**

Links

Research Compendium

The data and materials for this study are available on OSF (<https://osf.io/uenr7>) and GitHub (<https://github.com/jeksterslab/manMetaVAR>, <https://jeksterslab.github.io/manMetaVAR/index.html>).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` R package are available on GitHub (<https://github.com/jeksterslab/fitVARMxID>, <https://jeksterslab.github.io/fitVARMxID/index.html>).

metaDyn R Package

Source code and documentation for the `metaDyn` R package are available on GitHub (<https://github.com/jeksterslab/metaDyn>, <https://jeksterslab.github.io/metaDyn/index.html>).

Single Replication from the Simulation Study

This vignette presents a complete walk-through of one representative Monte Carlo replication, including data generation, estimation for each competing method, and selected output and diagnostics. Convergence diagnostics for a representative BMLVAR (Bayesian multilevel vector autoregressive model) replication, including PSR, ESS, and trace plots, are provided via the `Mplus` output section of the simulation-replication vignette.

<https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html>

Empirical Illustration: Meta-Analysis of Affect Dynamics

This vignette reproduces the empirical example reported in the manuscript, including data preprocessing, person-specific VAR estimation, second-stage meta-analytic synthesis, and the resulting population-level affect-dynamics summaries.

<https://jeksterslab.github.io/manMetaVAR/articles/empirical.html>

Benchmarking MetaVAR Against an Uncertainty-Uncorrected Two-Stage Approach and Bayesian Multilevel VAR

This vignette reports the benchmarking analysis comparing the computational demands of MetaVAR, the uncertainty-uncorrected two-stage benchmark, and BMLVAR under a common simulation setting.

<https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html>

Containers for Reproducibility

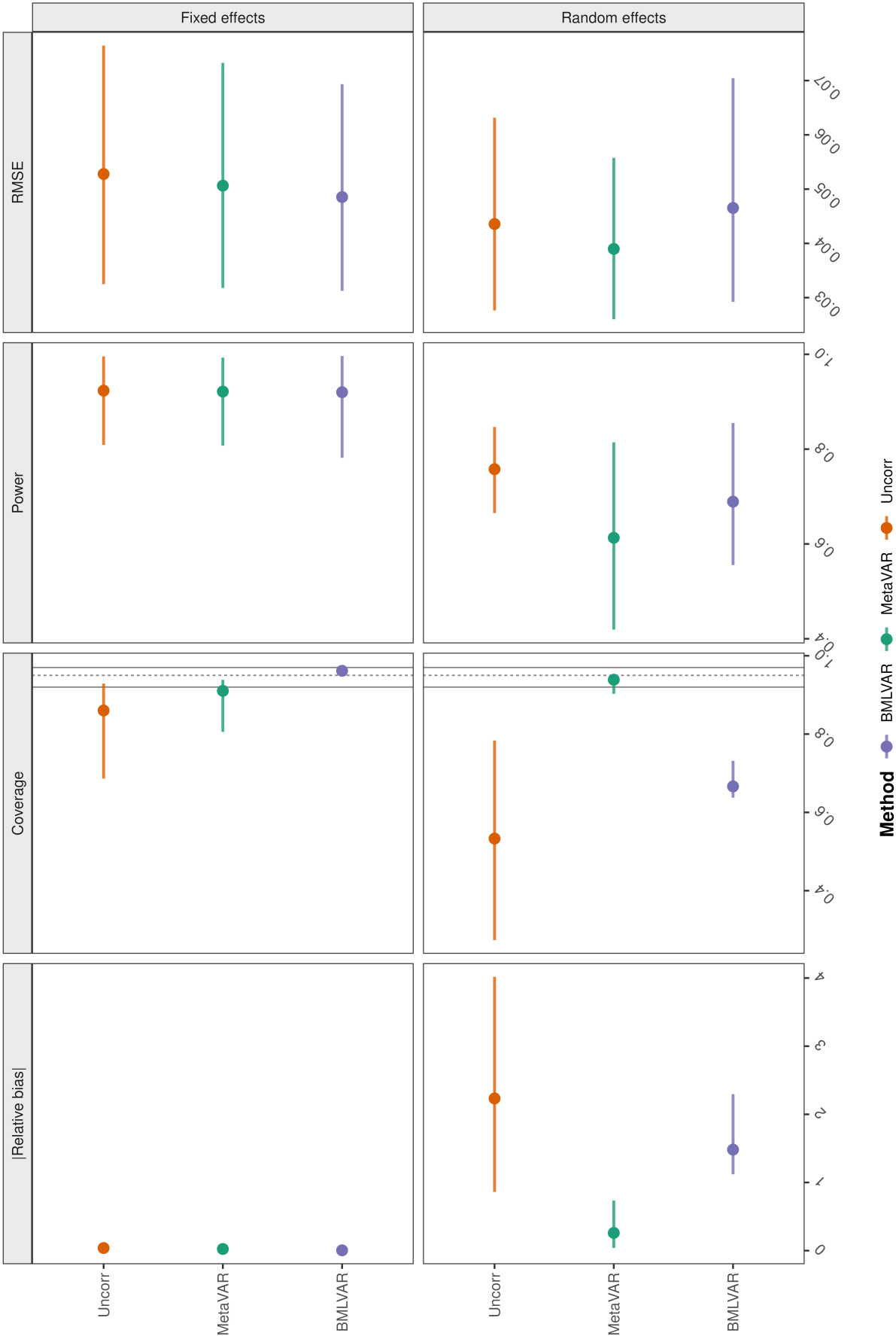
This vignette provides the containerized computing environments and step-by-step instructions needed to reproduce the analyses with the required software stack and dependencies.

<https://jeksterslab.github.io/manMetaVAR/articles/containers.html>

Figure 1
Overall Performance

Overall method performance across simulation design cells

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.



Coverage heatmap by parameter, design cell, and method

Negative values indicate undercoverage; zero indicates nominal coverage.

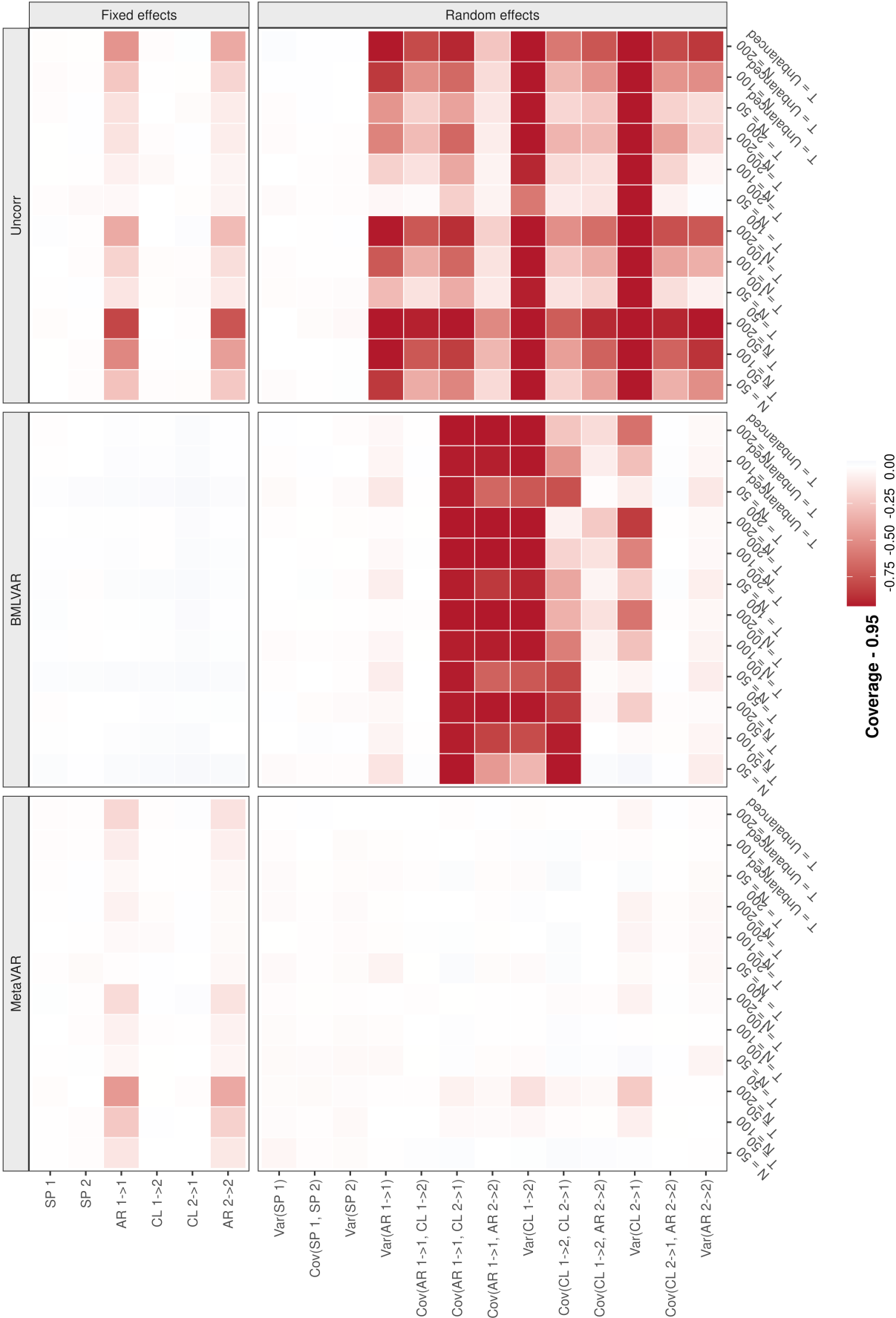


Figure 2
Coverage

Absolute value of the relative bias heatmap by parameter, design cell, and method

Smaller values are better. The heatmap shows magnitude only, not direction.

RMSE heatmap by parameter, design cell, and method

Figure 1 displays heatmaps of the Root Mean Square Error (RMSE) for fixed and random effects across different models and parameters. The figure is organized into a 3x2 grid of heatmaps. The columns represent 'Fixed effects' and 'Random effects'. The rows represent different models: 'Uncorr', 'BMLVAR', and 'MetaVAR'. The x-axis for all heatmaps represents parameters: SP 1, SP 2, AR 1->1, CL 1->2, CL 2->1, AR 2->2, Var(SP 1), Cov(SP 1, SP 2), Var(SP 2), Var(AR 1->1), CL 1->2, Cov(AR 1->1, CL 1->2), CL 2->1, Cov(AR 1->1, CL 2->1), AR 2->2, Cov(AR 1->1, AR 2->2), Var(CL 1->2), CL 1->2, CL 2->1, Cov(CL 1->2, CL 2->1), AR 2->2, Cov(CL 1->2, AR 2->2), Var(CL 2->1), CL 2->1, AR 2->2, Cov(CL 2->1, AR 2->2), and Var(AR 2->2). The y-axis for all heatmaps represents the number of observations (N) and the number of time points (T): N = 50, N = 100, N = 200, T = 50, T = 100, T = 200, T = Unbalanced. The color scale indicates the RMSE, with darker blue representing lower RMSE and lighter blue representing higher RMSE. The heatmaps show that RMSE generally decreases as N and T increase. The 'MetaVAR' model consistently shows the lowest RMSE across all parameters and conditions. The 'Uncorr' model shows the highest RMSE, particularly for the random effects parameters. The 'BMLVAR' model shows intermediate RMSE values. The 'Fixed effects' heatmaps show lower RMSE for the fixed effects parameters compared to the 'Random effects' heatmaps. The 'Random effects' heatmaps show higher RMSE for the random effects parameters, particularly for the variance and covariance parameters. The 'MetaVAR' model shows the lowest RMSE for all parameters, indicating superior performance in estimating both fixed and random effects.

Smaller values are better. RMSE is on the original parameter scale, so comparisons are most meaningful within parameter.

Power heatmap by parameter, design cell, and method

